

Actor-Critic Derivation

Remember,

$$\nabla J(\theta) = E_{\tau \sim \pi_\theta} \left[\sum_t Q_\pi(s_t, a_t) \nabla_\theta \log(\pi_\theta(a_t | s_t)) \right]$$

where $Q_\pi(s_t, a_t) = E[G_t | s_t, a_t]$

REINFORCE used G_t , now we use its expectation (expected reward for taking action a_t at s_t)

Define $V_\pi(s) = E_{a \sim \pi_\theta(a|s)} [Q_\pi(s, a)] = E_{a \sim \pi_\theta(a|s)} [E[G_t | s, a]]$

↓
Value function = expected reward at state s regardless of action

To evaluate how "good" an action is at state s , using its expected return $Q(s, a)$ be misleading because a high expected return doesn't tell us if the action is good. The state could just have a lot of actions with high return. Therefore, we should evaluate an action based on how good its expected return is based on the expected return across all actions at state s .

How good is the expected return of action a at state s ? $Q_\pi(s, a)$

How good is the expected return across all actions at state s ? $V_\pi(s)$.

Therefore, we should evaluate $Q_\pi(s, a) - V_\pi(s)$

Ex. At state s , we normally expect a return of 5. ($V_\pi(s) = 5$)

But for action a_1 , we expect a return of 3. ($Q_\pi(s, a_1) = 3$).

Previously, we would have used $Q_\pi(s, a_1) = 3$ to bump up $\pi(a_1 | s)$.

But, a_1 is actually bad compared to average so we use $Q_\pi(a_1, s) - V_\pi(s) = -2$ to decrease $\pi(a_1 | s)$.

Let $Q_\pi(s_t, a_t) = (Q_\pi(s_t, a_t) - V_\pi(s_t)) + V_\pi(s_t)$.

So, $\nabla J(\theta) = E_{\tau \sim \pi_\theta} \left[\sum_t [(Q_\pi(s_t, a_t) - V_\pi(s_t)) + V_\pi(s_t)] \cdot \nabla_\theta \log(\pi_\theta(a_t | s_t)) \right]$

$$= E_{\tau \sim \pi_\theta} \left[\sum_t [Q_\pi(s_t, a_t) - V_\pi(s_t)] \nabla_\theta \log(\pi_\theta(a_t | s_t)) \right] + E_{\tau \sim \pi_\theta} \left[\sum_t V_\pi(s_t) \nabla_\theta \log(\pi_\theta(a_t | s_t)) \right]$$

$$= \quad + E_{s_t \sim p(s_t | s_{t-1}, a_t), a_t \sim \pi_\theta(a_t | s_t)} [V_\pi(s_t) E[\nabla_\theta \log(\pi_\theta(a_t | s_t))]]$$

$$= \quad + E_{s_t} [V_\pi(s_t) \sum_{a_t} \pi_\theta(a_t | s_t) \frac{\nabla_\theta \pi_\theta(a_t | s_t)}{\pi_\theta(a_t | s_t)}]$$

$$= \quad + E_{s_t} [V_\pi(s_t) \nabla_\theta \underbrace{\sum_{a_t} \pi_\theta(a_t | s_t)}_{=1}]$$

$$= E_{\tau \sim \pi_\theta} \left[\sum_t \underbrace{(Q_\pi(s_t, a_t) - V_\pi(s_t))}_{= A_\pi(s_t, a_t) \text{ AKA "Advantage"}} \nabla_\theta \log(\pi_\theta(a_t | s_t)) \right] + 0$$

$= A_\pi(s_t, a_t)$ AKA "Advantage"

We want $A_\pi(s, a) = Q_\pi(s, a) - V_\pi(s)$, AKA, how much more expected reward do we get by taking action a at state s versus the expected reward over all actions at s ?

How to find $V_\pi(s) = E_{a \sim \pi(a|s)}[Q(s, a)]$? Let's learn $V_\phi(s) \approx V_\pi(s)$, the Critic

How to find $Q_\pi(s_t, a_t) = E_{\tau \sim \pi}[G_t | s_t, a_t]$?

LOIE:

$$= E_{\tau \sim \pi}[r_t + \gamma G_{t+1} | s_t, a_t]$$

$$E[X] = E_Y[E[X|Y]]$$

$$\text{By LOIE} = E_{\tau \sim \pi}[r_t | s_t, a_t] + \gamma E_{\tau \sim \pi}[G_{t+1} | s_t, a_t]$$

$$= E_{\tau \sim \pi}[r_t | s_t, a_t] + \gamma E_{s_{t+1}}[E_{\tau \sim \pi}[G_{t+1} | s_{t+1}] | s_t, a_t]$$

$$= E_{\tau \sim \pi}[G_{t+1} | s_{t+1}]$$

$$= E_{a_{t+1} \sim \pi(a|s)}[E_{\tau \sim \pi}[G_{t+1} | s_{t+1}]]$$

$$= E_{\tau \sim \pi}[r_t | s_t, a_t] + \gamma E_{s_{t+1}}[V_\pi(s_{t+1}) | s_t, a_t]$$

$$= E_{a_{t+1} \sim \pi(a|s)}[Q(s_{t+1}, a_{t+1})]$$

$$= E[r_t + \gamma \underbrace{V_\pi(s_{t+1})}_{\text{learned by critic}} | s_t, a_t]$$

$$= V_\pi(s_{t+1})$$

$$= E[r_t + \gamma V_\phi(s_{t+1}) | s_t, a_t]$$

$$\text{Fix one } t: \underbrace{r_t}_{\text{known through env.}} + \gamma \underbrace{V_\phi(s_{t+1})}_{\text{learned by critic, constant here}} = \hat{Q}_t(s_t, a_t)$$

In reinforce, we were approximating Q with $\hat{Q} = G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$

In actor-critic, we are approximating Q with $\hat{Q} = r_t + \gamma V_\phi(s_{t+1})$

$$\text{so, } \hat{A}_\pi(s, a) = \hat{Q}_\pi(s, a) - \hat{V}_\pi(s) = \underline{r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)}$$

$$\text{Therefore, } \nabla_\theta J(\theta) = E_{\tau \sim \pi} \left[\sum_t (r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)) \nabla_\theta \log(\pi_\theta(a_t | s_t)) \right] \approx E \left[\sum_t \delta_t \nabla_\theta \log(\pi_\theta(a_t | s_t)) \right]$$

During training, during step t when you have s_t ,

① Sample $a_t \sim \pi_\theta(a_t | s_t)$, get $V_\phi(s_t)$ from critic.

② Get r_t, s_{t+1} from $P(s_{t+1}, r_t | s_t, a_t)$

③ Get $V_\phi(s_{t+1})$ from critic.

④ Calculate $\hat{A}(s_t, a_t) = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) = \delta_t$

⑤ Update critic with $L_c = (\underbrace{r_t + \gamma V_\phi(s_{t+1})}_{\text{constant/no grad}} - V_\phi(s_t))^2$

⑥ Update actor like in REINFORCE with $L_A = - \underbrace{\delta_t}_{\text{no grad}} \nabla_\theta \log(\pi_\theta(a_t | s_t))$